

CSE 112:

Computer Organization
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End Semester Question Paper

IIIT-Delhi, Winter 2025

Duration : 120 Minutes

Date: Apr 28, 2025

Points: 80

Guidelines

If you found any ambiguity in any of the questions or there appears to be a lack of information, then write an **logical assumption** on the answer sheet to explain your side interpretation of the problem and solve accordingly.

First Block Instructions	Description	Operation
addi rd,rs, imm[11:0]	Add immediate	$rd = rs + sext(imm[11:0])$
sub rd, rs1, rs2	Subtract registers	$rd = rs1 - rs2$
add rd, rs1, rs2	Add registers	$rd = rs1 + rs2$
xor rd, rs1, rs2	Perform XOR boolean operation	$rd = rs1 \text{ xor } rs2$
slt rd, rs1, rs2	Set Less than	$rd = 1$. If $signed(rs1) < signed(rs2)$ else $rd=0$
sltu rd,rs1,rs2	Set Less than unsigned	$rd = 1$. If $unsigned(rs1) < unsigned(rs2)$ else $rd=0$
lw rd,#imm[11:0](rs1)	Load Word	$rd = mem(sext(rs1+imm))$
sw rd,#imm[11:0](rs1)	Store Word	$mem(sext(rs1+imm)) = rd$
beq rs1, rs2, imm[12:1]	Branch if equal	$PC = PC + sext(\{imm[12:1], 1'b0\})$ if $rs1 == rs2$
jal rd, imm[20:1]	Jump and Link	$rd = PC+4$. $PC = PC + sext(\{imm[20:1], 1'b0\})$
beq x0,x0,#0	Branch if equal	HALT

Second Block Instructions	Description	Operation
Jalr ra, rs1, imm[11:0]	Jump and Link register	$ra = PC+4$, $PC = rs1 + sext(imm[11:0])$

[CO2, CO3] Problem 1 : Pipeline**[13 Points]**

Draw the pipeline diagram of the below mentioned assembly program executed on a five stage risc-v processor with the microarchitecture described in figure[1]. You need to stall the pipeline in case of any hazard. Assume that there is always a cache hit and the instruction cache is different from the data cache.

Assembly Program

```

Addi x2,x0,#2
Addi x1,x0,#1
Addi x3,x0,#256
Lw x3,8(x3)
Add x4,x3,x1
Bne x2,x1,NEXT_1
Sub x3,x2,x1
Sub x3,x3,x0
NEXT_1: Addi x1,x0,#2
        Beq x1,x2,NEXT_2
        Sub x4,x5,x6
        addi x6,x0,#6
NEXT_2: Beq x0,x0,#0

```

[CO3] Problem II : Microprocessor Microarchitecture**[11+5 Points]**

You need to draw the pipeline diagram and stall the pipeline in case of unresolvable hazards, when the below-mentioned assembly code is executed on the microprocessor having microarchitecture as shown in fig[1]. And also, at each positive edge(posedge) of the clock, you need to write the status(in binary) of mentioned control signals of the microprocessor while executing the program.

A Hypothetical Example of Status Signals (for layout reference)

Posedge	StallIF	StallID	FlushD	FlushE	ForwardAE[1:0]	ALUSrcE
1st	1	0	0	0	01	0

Assembly Program

```
Addi x2,x0,#2
Addi x1,x0,#2
Addi x8,x0,#256
Lw x4,8(x8)
Add x5,x4,x1
Add x6,x5,x1
Beq x1,x2,LABEL
Add x7,x1,x2
Add x7,x1,x2
Sub x7, x2,x1
Beq x0,x0,#0
```

LABEL:

[CO1] Problem III : ISA Format**[01+01+01+02 Points]**

The format of an ISA is described below.. The ISA has 32 registers and each instruction size is 32-bit. A 32-bit processor follows this ISA and supports byte addressable memory access.

Opcode	rd	rs1	rs2
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You need to answer the below-mentioned questions

- How many bits are required to represent each register.
- How many number of bits are required to represent each opcode
- How many uniques instructions can the ISA support
- What decimal number should be added to the Program Counter to fetch the next instruction.

[CO2] Problem IV : Assembly Encoding**[10 Points]**

Write an assembly program to print the below-mentioned pattern.

Out rs1	Print the content of register rs1.
Out "\n"	Go to next line.

Pattern

1 2 3 4 5

1 2 3 4

1 2 3

1 2

1

[CO4] Problem V : Cache Indexing**[03*3+01 Points]**

A processor having byte addressable range of 64GByte is connected to a 12-way set associative cache. The size of the block in cache is 64 Bytes and the cache size is 48KByte.

- Find the number of bits in tag, set index, and block offset
- For this cache system, if you were to place block number 64 (assume block numbers start from 0) from main memory to the cache. Which set number (assuming set number counting starts from 0) will you put it in?

[CO4] Problem VI : Cache Miss Rate**[05 Points]**

Suppose you have a cache with a hit latency of 1 cycle. When it misses, it goes to main memory, which has a latency of 105 cycles (this includes the latency required to know that the access was a miss and the latency to bring the data from main memory to cpu). If the average memory access time (in cycles) is 6.2 cycles. What is the miss rate of the cache?

[CO1, CO3] Problem VII : Processor Speedup**[05 Points]**

You have built a non-pipelined processor with a frequency of 2.5 gigahertz, and the average cycles per instruction is 4. The same processor is upgraded to a pipelined processor with five stages, but due to the internal pipeline delay, the clock speed is reduced to 2 gigahertz. Assume there are no stalls in the pipeline. What is the ratio of speedup of the non-pipelined processor to the speedup of the pipeline processor for the execution of 100 instructions?

[CO2] Problem VIII : Caller Callee**[10 Points]**

You need to convert the below-mentioned pseudo program into an assembly language program. The isa supports **First Block and Second Block instructions** as specified in the beginning of the paper.

- The ISA has 32 general purpose registers x0,x1,...,x31
- x1 is the return address register
- x10, x11 stores the return value from the function.
- The arguments to the callee are stored in registers x12, x13, ..., x17.
- The registers from x18, x19, ..., x27 are caller saved.
- The registers x8, x9 are callee saved.
- Whenever the branch and link instruction is used. The return address is stored in x1, and the program counter jumps to the specified label.

Pseudo Program

```
Void func1() {
    a=5
    b=9
    c = 11
    d=func2(a,b,c)
    f=d+a
}
Int func2(a,b,c) {
    return a+b-c
}
```

[CO3] Problem IX: Branch Penalty**[06 Points]**

Consider two processors C, D following RISC ISA and having stage description as follows..

- Processor C has 36 stages as F1, F2, ... F10, D1, D2, ... D9, E1, E2, ... E9, M1, M2, ... M4, W1, W2, ... W4. The branch operations are determined in the stage E6. The processor operates at 1GHz.
- Processor D has 92 stages as F1, F2, ... F30, D1, D2, ... D10, E1, E2, ... E20, M1, M2, ... M25, W1, W2, ... W7. And, the branch operations are determined in the stage E6. The processor operates at 5GHz.

You need to calculate the Branch penalty of both the processors

- In terms of number of cycles.
- In terms of time lost in the penalty in ns.

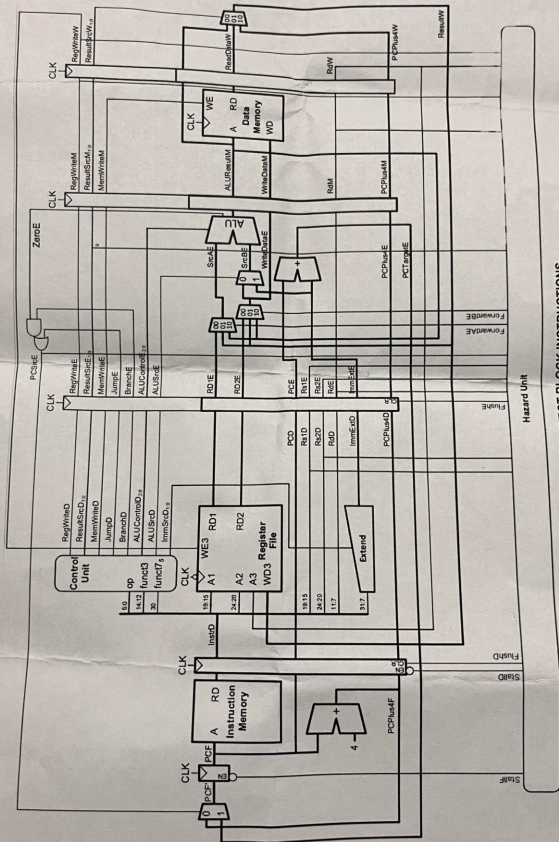


Figure 1: Microarchitecture of Five Stage RISC-V Processor following FIRST BLOCK INSTRUCTIONS.
 Image Source: S. L. Harris and D. Harris, "Digital Design and RISC-V Computer Architecture Textbook," 2021 ACM/IEEE Workshop on Computer Architecture Education (WCAE), Raleigh, NC, USA, 2021